Weight Initialization: Xavier vs. He Mechanisms

Theoretical Reference for MLP Defense (Soutenance)

Multilayer Perceptron Project

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Table des matières

1 Introduction to the Variance Problem

English : The goal of weight initialization is to prevent the variance of the activations and gradients from exploding or vanishing as they pass through the layers of a deep neural network. If we denote n_{in} as the number of inputs to a neuron, the total pre-activation is $z = \sum_{i=1}^{n_{in}} W_i a_i$. Assuming independent and identically distributed variables with zero mean, the variance propagates as :

$$\mathrm{Var}(z) = n_{in} \cdot \mathrm{Var}(W) \cdot \mathrm{Var}(a)$$

To preserve the signal flowing forward and backward, we want $\mathrm{Var}(z) = \mathrm{Var}(a)$, which implies that we must calibrate $\mathrm{Var}(W) \approx \frac{1}{n_{in}}$.

Español : El objetivo de la inicialización de pesos es evitar que la varianza de las activaciones y los gradientes explote o se desvanezca al avanzar por las capas. Si denotamos n_{in} como el número de entradas de una neurona, la preactivación total es $z = \sum_{i=1}^{n_{in}} W_i a_i$. Asumiendo variables independientes e idénticamente distribuidas con media cero, la varianza se propaga como $\mathrm{Var}(z) = n_{in} \cdot \mathrm{Var}(W) \cdot \mathrm{Var}(a)$. Para preservar la señal, buscamos que $\mathrm{Var}(z) = \mathrm{Var}(a)$, lo que implica calibrar la varianza de los pesos de modo que $\mathrm{Var}(W) \approx \frac{1}{n_{in}}$.

Français : L'objectif de l'initialisation des poids est d'empêcher la variance des activations et des gradients d'exploser ou de s'évanouir à travers les couches. Si l'on note n_{in} le nombre d'entrées d'un neurone, la pré-activation totale est $z = \sum_{i=1}^{n_{in}} W_i a_i$. En supposant des variables indépendantes et identiquement distribuées de moyenne nulle, la variance se propage comme $\mathrm{Var}(z) = n_{in} \cdot \mathrm{Var}(W) \cdot \mathrm{Var}(a)$. Pour préserver le signal, nous voulons que $\mathrm{Var}(z) = \mathrm{Var}(a)$, ce qui implique de calibrer la variance des poids de sorte que $\mathrm{Var}(W) \approx \frac{1}{n_{in}}$.

2 Xavier / Glorot Initialization (Sigmoid & Tanh)

Mathematical Derivation of Limits

Xavier initialization targets symmetric activations centered at zero. The derived optimal variance is :

$$\mathrm{Var}(W) = \frac{2}{n_{in} + n_{out}}$$

If we use a Uniform Distribution $U(-r, r)$, the mathematical variance of any continuous uniform distribution is $\mathrm{Var}(U) = \frac{r^2}{3}$. Matching both equations :

$$\frac{r^2}{3} = \frac{2}{n_{in} + n_{out}} \implies r = \sqrt{\frac{6}{n_{in} + n_{out}}}$$

Descriptions

English : Designed strictly for linear or symmetric activations like Sigmoid and Hyperbolic Tangent (tanh). It assumes the activation function acts linearly near the origin.

Español : Diseñada estrictamente para activaciones simétricas o lineales en el origen como la Sigmoide y la Tangente Hiperbólica (tanh). Asume que la función de activación se comporta de forma lineal cerca del origen.

Français : Conçue strictement pour les activations symétriques ou linéaires à l'origine comme la Sigmoïde et la Tangente Hyperbolique (tanh). Elle suppose que la fonction d'activation se comporte de manière linéaire près de l'origine.

3 He / Kaiming Initialization (ReLU)

Mathematical Derivation of Limits

When using ReLU, negative values are completely zeroed out. Geometrically, this drops exactly half of the variance signal : $\text{Var}(a) = \frac{1}{2}\text{Var}(z)$. To compensate for this 50% loss of energy per layer, He doubled the target variance :

$$\text{Var}(W) = \frac{2}{n_{in}}$$

If we map this to a Uniform Distribution $U(-r, r)$ where $\text{Var}(W) = \frac{r^2}{3}$:

$$\frac{r^2}{3} = \frac{2}{n_{in}} \implies r = \sqrt{\frac{6}{n_{in}}}$$

Descriptions

English : Essential for Rectified Linear Units (ReLU) and its variants. It accounts for the mathematical reality that ReLU rectifies half of the input space to zero.

Español : Esencial para unidades lineales rectificadas (ReLU) y sus variantes. Resuelve la realidad matemática de que ReLU apaga la mitad del espacio de entrada a cero.

Français : Essentielle pour les unités linéaires rectifiées (ReLU) et ses variantes. Elle compense le fait mathématique que ReLU annule la moitié de l'espace d'entrée.

4 Consequences of Misconfiguration

Choosing the wrong initialization pair triggers destructive numerical phenomena during the training loop :

1. Vanishing Gradient / Gradiente Desvanecido / Gradient Évanouissant

- **Scenario :** Using Xavier with a very deep ReLU network, or initializing with weights that are too small.
- **Result :** The variance of the signal shrinks exponentially layer after layer. The gradients flowing backwards drop to zero. The lower layers of the MLP stop updating their weights entirely.

2. Exploding Gradient / Gradiente Explosivo / Gradient Explosif

- **Scenario :** Initializing a Sigmoid/Tanh network with He parameters, or setting initial weights that are too large.
- **Result :** The variance grows exponentially with network depth. For Sigmoid/Tanh, large pre-activations push the neurons into the flat saturated zones, killing the gradient. For unbounded activations, values turn into NaN within a few epochs.

3. Dead Neurons / Neuronas Muertas / Neurones Morts

- **Scenario :** Misconfiguring He initialization on a ReLU network.
- **Result :** If weights are initialized too negatively, neurons output 0 for the entire dataset. Since the derivative of ReLU is 0 for negative inputs, no gradient updates them, creating permanently inactive nodes.

5 Quick Reference Compatibility Matrix

Activation	**Ideal Initialization**	**Target Variance**	**Uniform Limits**
Linear	Xavier / Glorot	$\frac{2}{n_{in}+n_{out}}$	$\sqrt{\frac{6}{n_{in}+n_{out}}}$
Sigmoid	Xavier / Glorot	$\frac{2}{n_{in}+n_{out}}$	$\sqrt{\frac{6}{n_{in}+n_{out}}}$
Tanh	Xavier / Glorot	$\frac{2}{n_{in}+n_{out}}$	$\sqrt{\frac{6}{n_{in}+n_{out}}}$
ReLU	He / Kaiming	$\frac{2}{n_{in}}$	$\sqrt{\frac{6}{n_{in}}}$